



UBER DATA HIDDEN PATTERNS

Data analysis



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UNDERCOVER DETECTIVES:

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Introduction:

Uber is a global ride-hailing platform that connects riders with drivers through a mobile app. Understanding how Uber trips vary by time, vehicle type, payment method, and other factors can help improve service quality and operational decisions.

In this project, we analyze the [Uber Ride Analytics Dashboard dataset](#), which contains real ride-sharing data with details about bookings, trip status, revenue, cancellations, vehicle types, and ratings. Our goal is to explore patterns and trends in the data, such as booking behavior, cancellation reasons, and factors that influence ride experience and booking value.

This analysis will uncover useful insights about Uber's operations and help support better decision-making in pricing, customer service, and ride planning.

Dataset Overview:

The **Uber Ride Analytics Dashboard dataset** contains detailed information about Uber trips, including completed, cancelled, and incomplete rides. It is a rich dataset suitable for analyzing booking patterns, cancellations, revenue, and ride experience.

The dataset includes the following important features:

- **Date & Time** – when the booking was made.
- **Booking ID** – unique identifier for each ride.
- **Booking Status** – whether the ride was completed, cancelled by customer, or cancelled by driver.
- **Vehicle Type** – type of vehicle used (Go Mini, Go Sedan, Auto, eBike/Bike, UberXL, Premier Sedan).
- **Pickup & Drop Location** – starting and ending points of the ride.
- **Avg VTAT** – average time for the driver to reach the pickup location (in minutes).
- **Avg CTAT** – average trip duration from pickup to drop location (in minutes).
- **Cancelled Rides & Reasons** – flags and reasons for cancellations by customer or driver.
- **Incomplete Rides & Reasons** – flags and reasons for rides that were not completed.
- **Booking Value** – total fare amount for the ride.
- **Ride Distance** – distance covered during the ride (in km).
- **Driver & Customer Ratings** – feedback ratings for driver and customer (1-5 scale).
- **Payment Method** – payment type used (UPI, Cash, Credit Card, Uber Wallet, Debit Card).

This dataset allows for detailed exploratory data analysis (EDA) to find patterns in ride bookings, cancellations, fare amounts, and overall service quality.

Data Preprocessing & Cleaning :

- **Initial Data Exploration**
The dataset was explored to understand its structure, data types, and overall quality.
- **Duplicate Records Check**
The data was checked for duplicate records, and no duplicates were found.

```
|: Uber_df.duplicated().sum()  
|: 0
```

- **Handling Missing Values**
Missing values were handled based on their proportion and data type:
- Numerical features were filled using **median values**.
- Categorical features were filled using **mode** or labeled categories.
- Indicator flags were used where needed to preserve information.

```

# Driver Cancellation Reason
In [20]: high_cat_missing = ['Incomplete Rides Reason', 'Reason for cancelling by Customer', 'Driver Cancellation Reason']
        for col in high_cat_missing:
            Uber_clean[col] = np.where(
                Uber_clean['Booking Status'] == 'Completed',
                Uber_clean[col],
                Uber_clean[col].fillna('Unknown'))

In [21]: moderate_num_missing = missing_data[
        missing_data.apply(
            lambda row: 30 <= row['Percentage_of_missing_data'] < 70 and row['types'] == 'Float64',
            axis=0
        )
        ].index.tolist()

In [22]: moderate_num_missing
Out[22]: ['Customer Rating',
         'Driver Ratings',
         'Ride Distance',
         'Booking Value',
         'Avg CTAT']

In [23]: moderate_num_missing['Customer Rating', 'Driver Ratings', 'Ride Distance', 'Booking Value', 'Avg CTAT']
        for col in moderate_num_missing:
            Uber_clean[col] = Uber_clean[col].fillna(Uber_clean[col].median())

In [24]: moderate_cat_missing = missing_data[
        missing_data.apply(
            lambda row: 30 <= row['Percentage_of_missing_data'] < 70 and row['types'] == 'object',
            axis=0
        )
        ].index.tolist()

In [25]: moderate_cat_missing
Out[25]: ['Payment Method']

In [26]: moderate_cat_missing['Payment Method']
        for col in moderate_cat_missing:
            Uber_clean[col] = Uber_clean[col].fillna(Uber_clean[col].mode()[0])

In [27]: Uber_clean['Avg VTAT'].fillna(Uber_clean['Avg VTAT'].median(), inplace=True)

```

- **Date and Time Processing**

The Date and Time columns were merged into a single **datetime** feature.

```

In [33]: # We noticed the date & time columns are string so it's best to convert it to DateTime and merge it together
        Uber_clean['DateTime'] = pd.to_datetime(Uber_clean['Date'] + ' ' + Uber_clean['Time'])

In [34]: Uber_clean[['DateTime']]
Out[34]:
           DateTime
0  2024-03-23 12:29:38

```

- **Feature Engineering**

Additional time-based features were created to support deeper analysis, including:

- Hour
- Day of Week
- Month
- Peak Hour
- Time of Day

- **Data Cleaning**

Unnecessary characters were removed to ensure consistency and uniform formatting across the dataset

- Check our [notebook file](#)

for better analysis we need to calculate Hour, DayOfWeek, Month

```

In [35]: Uber_clean['Hour'] = Uber_clean['DateTime'].dt.hour
        Uber_clean['DayOfWeek'] = Uber_clean['DateTime'].dt.day_name()
        Uber_clean['Month'] = Uber_clean['DateTime'].dt.month_name()

In [36]: Uber_clean
Out[36]:

```

```

In [37]: def peak_hour(hour):
        if hour > 7 and hour < 10:
            return 1
        elif hour > 17 and hour < 21:
            return 1
        else:
            return 0

In [38]: Uber_clean['Peak_hour'] = Uber_clean['Hour'].apply(peak_hour)

In [39]: Uber_clean['Peak_Hour_Label'] = Uber_clean['Peak_hour'].map({1: "Peak", 0: "Non-Peak"})

In [40]: def time_of_day(hour):
        if hour >= 6 and hour < 12:
            return "Morning"
        if hour > 12 and hour < 17:
            return "Afternoon"
        if hour >= 17 and hour < 21:
            return "Evening"
        else:
            return "Night"

In [41]: Uber_clean['Time_of_Day'] = Uber_clean['Hour'].apply(time_of_day)

In [42]: Uber_clean['Time_of_Day'].value_counts()
Out[42]:
Time_of_Day
Evening    44118
Morning    42672
Night      32874
Afternoon  30336
Name: Time_of_Day, dtype: int64

```

1. Overview of Bookings

The analysis shows that approximately **62% of total bookings result in completed trips**, indicating a generally good operational performance.

- **Driver cancellations** account for **18%** of bookings.
- **Customer cancellations** represent **7%**.
- Cases of **No Driver Found** and **incomplete trips** together make up around **13%** of total requests, which highlights an area that requires improvement.

Overall, the majority of requests are successfully fulfilled, but operational inefficiencies still exist.

2. Temporal Analysis (Monthly – Daily – Hourly)

Monthly trends:

The highest number of bookings occurs during the **last quarter of the year**, particularly from **October to December**, which also corresponds to the highest total booking value. This suggests strong seasonal demand toward the end of the year.

Weekly trends:

- **Friday and Saturday** show the highest demand levels.
- **Midweek days** experience relatively lower demand.

Hourly trends:

- **Evening hours** have the highest number of bookings.
- Followed by **morning**, then **afternoon**.
- **Late-night hours** record the lowest demand.



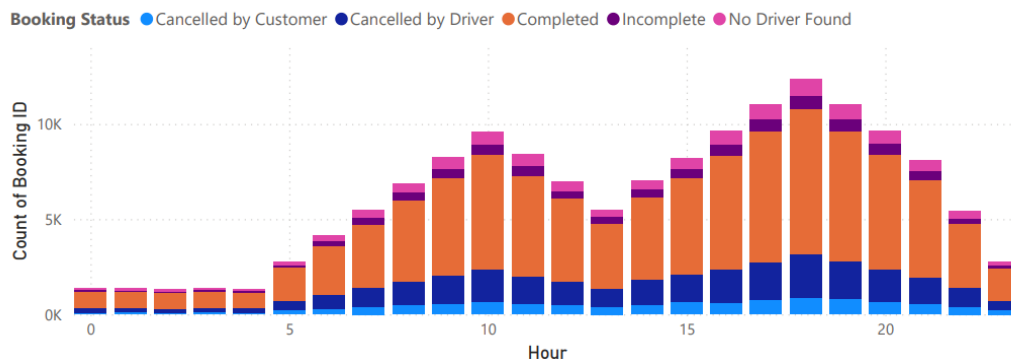
3. Booking Status and Peak Hours

During peak hours:

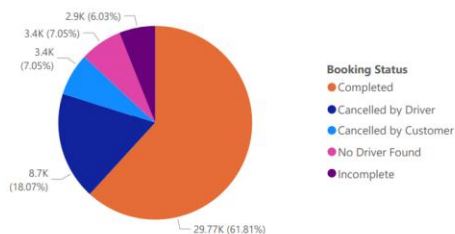
- More than **62% of trips are completed.**
- **Driver cancellations remain higher** than customer cancellations.
- Increased demand during peak periods raises the likelihood of cancellations or situations where no driver is available.

This indicates that peak-time pressure impacts service reliability, though completion rates remain relatively strong.

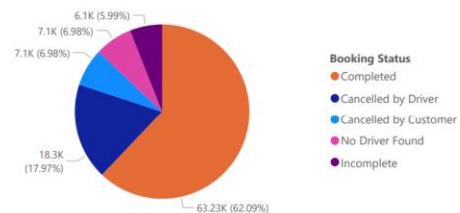
Count of Booking ID by Hour and Booking Status



Count of Booking Status by Booking Status and Booking Status



Count of Peak_Hour_Label by Booking Status and Booking Status



4. Customer Cancellation Reasons

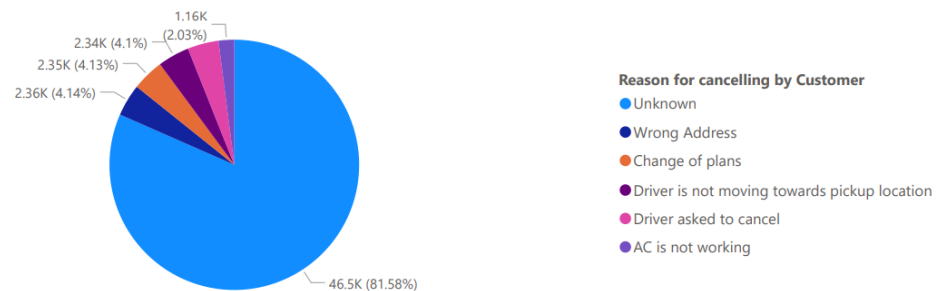
Over **80% of customer cancellations** are classified as **“Unknown”**, pointing to weak data collection regarding cancellation reasons.

Identified reasons include:

- Incorrect pickup address
- Change of plans
- Driver not heading toward the pickup location
- Driver requesting the customer to cancel

This highlights the need to improve both the **user experience** and the **cancellation reason documentation process**.

Count of Booking Status by Reason for cancelling by Customer



5. Driver Cancellation Reasons

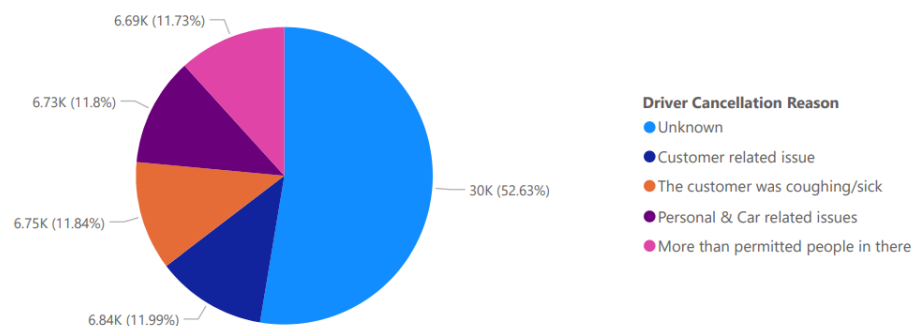
More than **half of driver cancellations** are also labeled as “**Unknown**”.

Other recurring reasons include:

- Issues related to the customer
- Personal or vehicle-related problems
- Number of passengers exceeding the allowed limit

This suggests a lack of structured reporting and monitoring of driver-side cancellations.

Count of Booking Status by Driver Cancellation Reason



6. Incomplete Trips

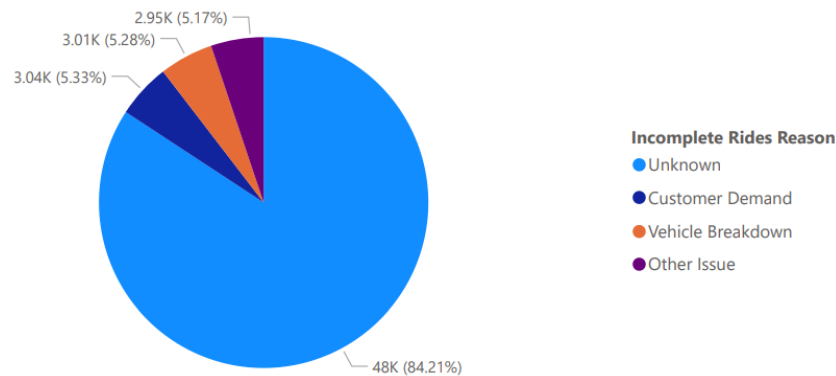
Approximately **84% of incomplete trips** have an **unknown cause**.

Other reported reasons include:

- High demand exceeding supply
- Vehicle breakdowns
- Other operational issues

This further emphasizes the importance of improving operational logging and transparency.

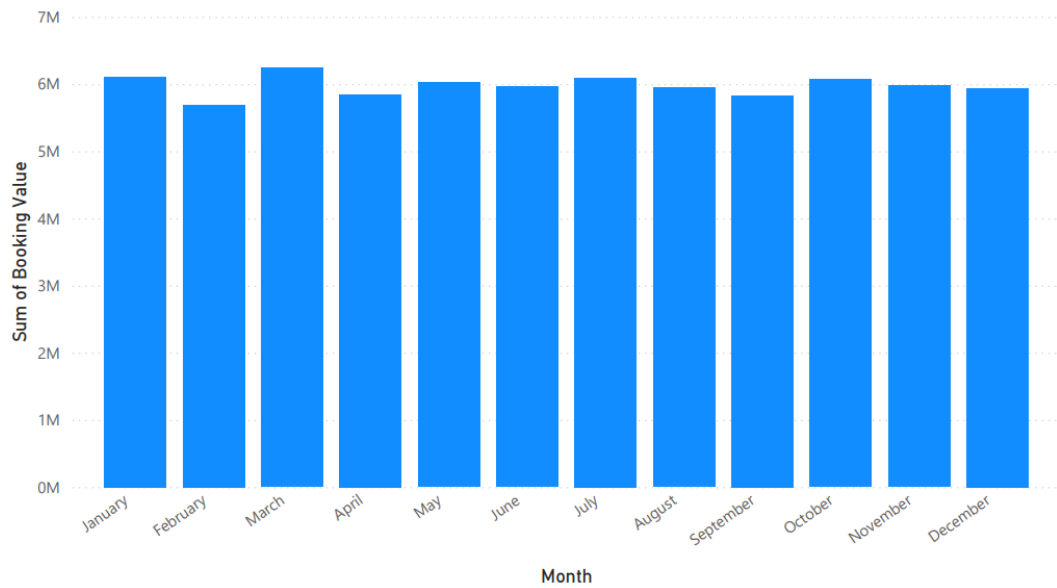
Count of Incomplete Rides Reason by Incomplete Rides Reason



7. Financial Analysis

The **total booking value increases gradually throughout the year**, reaching its peak in the **final quarter**. This reflects **seasonal demand growth** and an improvement in financial performance toward year-end.

Sum of Booking Value by Month



Is Uber a Successful and Sustainable Business?

Based on the data analysis, **Uber can be considered a successful and sustainable business rather than a short-term success**, for the following reasons:

1. Consistent Demand Throughout the Year

Bookings are recorded in all months without interruption, with a significant increase in demand and booking value toward the end of the year. This indicates long-term user reliance and service stability.

2. High Trip Completion Rate

Achieving a **62% trip completion rate** demonstrates Uber's ability to meet demand and operate at a reasonable efficiency level, which is a key indicator of operational success.

3. Strong Performance During Peak Hours

Despite increased pressure during peak periods, the majority of trips are still completed, reflecting system resilience and scalability.

4. Operational Challenges Are Manageable

Cancellations and "No Driver Found" issues are common challenges in ride-hailing platforms. However, they have not led to a decline in demand or revenue and can be addressed through better driver management, incentives, and improved user experience.

5. Revenue Growth Indicates Sustainability

The steady rise in total booking value especially in the last quarter suggests financial growth rather than decline, reinforcing the conclusion that Uber's success is **ongoing and sustainable**, not temporary.

Conclusion

In conclusion, the analysis of Uber's booking data indicates that the platform demonstrates **stable operational performance and noticeable financial growth**, particularly in the latter months of the year. Although certain operational challenges exist—such as relatively high cancellation rates and unclear reasons for incomplete rides—these issues do not represent indicators of failure. Rather, they highlight **opportunities for improvement and optimization**.

The findings confirm that Uber is **not a short-term success**, but rather a **successful and sustainable business model**, supported by consistent demand, a high proportion of completed rides, and the system's ability to handle peak-hour pressures effectively. Furthermore, enhancing the documentation of cancellation reasons, improving user experience, and increasing driver management efficiency would help reduce operational issues and improve customer satisfaction.

Based on these results, it can be concluded that Uber possesses the **operational and financial foundations necessary for long-term sustainability and growth**, with strong potential for improved performance through continuous, data-driven optimization.